

YOLOv11 vs YOLOv8 - Evaluation Report

Source files: `results.csv` (YOLOv11) and `results_analysis_values.csv` (YOLOv8). Both runs cover 50 epochs. Metrics are validation metrics for bounding boxes, taken from the training logs. F1-score is computed as $2 \cdot P \cdot R / (P + R)$ from the logged precision and recall.

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1. Accuracy comparison (final epoch, epoch 50)

Metric	YOLOv11	YOLOv8	Difference (v8 - v11)	Winner
Precision	0.4409	0.8988	+0.4579	YOLOv8
Recall	0.4264	0.8705	+0.4441	YOLOv8
F1-score	0.4335	0.8844	+0.4509	YOLOv8
mAP@50	0.3872	0.9234	+0.5362	YOLOv8
mAP@50-95	0.1440	0.6634	+0.5194	YOLOv8

YOLOv8 scores about 2.0× higher on precision, recall and F1, 2.4× higher on mAP@50 and 4.6× higher on mAP@50-95.

2. Best epoch per metric

Metric	YOLOv11 (best, epoch)	YOLOv8 (best, epoch)		
Precision	0.4568 (47)	0.9157 (43)		
Recall	0.4285 (49)	0.8895 (48)		
F1-score	0.4347 (47)	0.8844 (50)		
mAP@50	0.3914 (47)	0.9246 (43)		

Metric	YOLOv11 (best, epoch)	YOLOv8 (best, epoch)		
mAP@50-95	0.1451 (46)	0.6684 (41)		

3. Stability: mean of last 5 epochs (46–50)

Metric	YOLOv11	YOLOv8
Precision	0.4441	0.8776
Recall	0.4212	0.8681
F1-score	0.4323	0.8725
mAP@50	0.3879	0.9208
mAP@50-95	0.1437	0.6519

The last-5 means are close to the final-epoch values for both models, so the final numbers are not a lucky spike. YOLOv8 fluctuates more from epoch to epoch (for example, recall ranges from 0.82 to 0.89 over the final 10 epochs), while YOLOv11 rises smoothly.

4. Training progression (mAP@50 / mAP@50-95 / F1)

Epoch	YOLOv11 mAP@50	YOLOv11 mAP@50-95	YOLOv11 F1	YOLOv8 mAP@50	YOLOv8 mAP@50-95	YOLOv8 F1
1	0.109	0.034	0.185	0.676	0.400	0.662
10	0.181	0.062	0.255	0.862	0.579	0.812
20	0.297	0.108	0.347	0.856	0.597	0.803
30	0.330	0.125	0.375	0.903	0.639	0.844
40	0.366	0.136	0.411	0.921	0.654	0.865
50	0.387	0.144	0.434	0.923	0.663	0.884

YOLOv8 starts at 0.68 mAP@50 in epoch 1 and plateaus above 0.90 by epoch 30. YOLOv11 starts at 0.11, is still improving at epoch 50 (about +0.02 mAP@50 over the last 10 epochs) and has not converged.

5. Loss comparison (epoch 1 → epoch 50)

Loss	YOLOv11	YOLOv8
Train box	2.567 → 1.898	1.346 → 0.679
Train cls	2.892 → 1.634	2.148 → 0.411
Train DFL	1.958 → 1.533	1.359 → 0.968
Val box	2.458 → 2.178	1.330 → 0.968
Val cls	2.790 → 1.833	2.242 → 0.628
Val DFL	1.892 → 1.640	1.393 → 1.210

Both models' validation losses fall alongside training losses, so neither shows clear overfitting. YOLOv11's losses remain much higher at epoch 50, consistent with its lower accuracy.

6. Training time

	YOLOv11	YOLOv8
Total (50 epochs)	3,022 s (≈ 50 min)	1,272 s (≈ 21 min)
Avg per epoch	≈ 60 s	≈ 25 s

7. Conclusions

- On these logs, YOLOv8 is clearly the stronger model on every metric: precision, recall, F1, mAP@50 and mAP@50-95.
- The widest gap is mAP@50-95 (0.663 vs 0.144), meaning YOLOv11's boxes are also much less accurately localised at strict IoU thresholds.
- YOLOv11 is still improving at epoch 50, so it is under-trained relative to YOLOv8. Longer training would likely narrow the gap, but it is unlikely to close it within a few more epochs.

8. Caveats worth checking

A gap this large is unusual, since YOLOv11 normally matches or beats YOLOv8 on the same data. Before drawing conclusions about the architectures, please verify:

- Both runs used the same dataset, train/val split, image size, batch size and epochs.
- Both started from the same kind of weights. YOLOv8's epoch-1 mAP@50 of 0.68 suggests pretrained weights, while YOLOv11's 0.11 looks like training from scratch or a much weaker start.
- The two files are labelled correctly (results = YOLOv11, results analysis values = YOLOv8). This report uses your labels as given.
- Per-class metrics are not available in these files, so this report covers overall (all-class) metrics only.